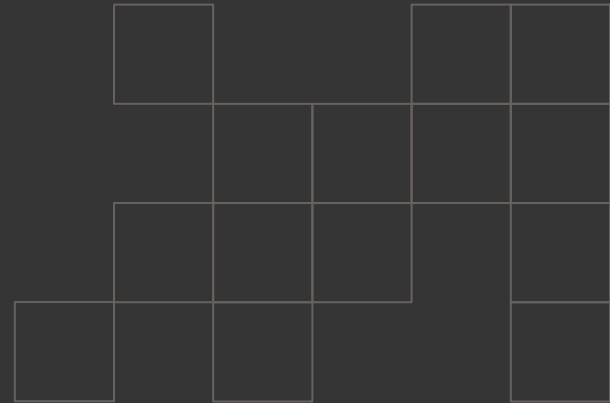


Database Normalization



Database normalization is the systematic process of organizing data in a relational database to reduce redundancy and improve data integrity by dividing large tables into smaller, related ones. Key rules include achieving 1NF (atomic values), 2NF (no partial dependencies), and 3NF (no transitive dependencies). **This minimizes anomalies (update, insert, delete)**, improves consistency, and optimizes query performance.



1NF, 2NF, 3NF

<https://www.freecodecamp.org/news/database-normalization-1nf-2nf-3nf-table-examples/>

Consider this:

CD Number	Band	Album	Track #	Tune	Composer	Band Name1	Band Name 2	Band Name 3	Band Name 4	DateRecorded	Date Released
43279	Beatles	White Album	1	Back in the USSR	McCartney	John Lennon	Paul McCartney	George Harrison	Ringo Starr	10/7/1968	11/22/1968
43279	Beatles	White Album	2	Dear Prudence	Lennon	John Lennon	Paul McCartney	George Harrison	Ringo Starr	10/15/1968	11/22/1968
43279	Beatles	White Album	3	Glass Onion	Lennon	John Lennon	Paul McCartney	George Harrison	Ringo Starr	11/2/1968	11/22/1968
43279	Beatles	White Album	4	Ob-La-Di, Ob-La-Da	McCartney	John Lennon	Paul McCartney	George Harrison	Ringo Starr	10/9/1968	11/22/1968
43279	Beatles	White Album	5	Wild Honey Pie	McCartney	John Lennon	Paul McCartney	George Harrison	Ringo Starr	10/25/1968	11/22/1968
43279	Beatles	White Album	6	Bungalow Bill	Lennon	John Lennon	Paul McCartney	George Harrison	Ringo Starr	10/1/1968	11/22/1968
43279	Beatles	White Album	7	While My Guitar Gently Weeps	Harrison	John Lennon	Paul McCartney	George Harrison	Ringo Starr	10/20/1968	11/22/1968
43279	Beatles	White Album	8	Happiness is a Warm Gun	Lennon	John Lennon	Paul McCartney	George Harrison	Ringo Starr	11/4/1968	11/22/1968

- The release date for *White Album* is listed as 11/22/1968 on every row.
If we discover the release date was entered incorrectly, how many rows must we update?
 - What happens if we forget one row?
- The band members (John Lennon, Paul McCartney, etc.) appear in every row. If Ringo's name were misspelled in one row, what problem would that create?
 - Could we end up with both "Ringo Starr" and "Ringo Star" in the same database?
- Suppose The Beatles record a new album, but no tracks have been recorded yet. Can we insert the album into this table?
 - Why or why not?
- If we delete the row for "Back in the USSR," what information do we lose?
- If we delete the last remaining track of the album, what happens to: Album information? Band information? Release date?
- Should deleting a song delete the album?

CD Number	Band	Album	Track #	Tune	Composer	Band Name1	Band Name 2	Band Name 3	Band Name 4	DateRecorded	Date Released
43279	Beatles	White Album	1	Back in the USSR	McCartney	John Lennon	Paul McCartney	George Harrison	Ringo Starr	10/7/1968	11/22/1968
43279	Beatles	White Album	2	Dear Prudence	Lennon	John Lennon	Paul McCartney	George Harrison	Ringo Starr	10/15/1968	11/22/1968
43279	Beatles	White Album	3	Glass Onion	Lennon	John Lennon	Paul McCartney	George Harrison	Ringo Starr	11/2/1968	11/22/1968
43279	Beatles	White Album	4	Ob-La-Di, Ob-La-Da	McCartney	John Lennon	Paul McCartney	George Harrison	Ringo Starr	10/9/1968	11/22/1968
43279	Beatles	White Album	5	Wild Honey Pie	McCartney	John Lennon	Paul McCartney	George Harrison	Ringo Starr	10/25/1968	11/22/1968
43279	Beatles	White Album	6	Bungalow Bill	Lennon	John Lennon	Paul McCartney	George Harrison	Ringo Starr	10/1/1968	11/22/1968
43279	Beatles	White Album	7	While My Guitar Gently Weeps	Harrison	John Lennon	Paul McCartney	George Harrison	Ringo Starr	10/20/1968	11/22/1968
43279	Beatles	White Album	8	Happiness is a Warm Gun	Lennon	John Lennon	Paul McCartney	George Harrison	Ringo Starr	11/4/1968	11/22/1968

What is the Primary Key?

- Is CD Number enough?
- Is Track # enough?
- Is Tune enough?
- Would it have to be a composite key?

What does one row represent?

- This table can work as a record or notation.
- But it doesn't scale.
- And databases are built to scale.

Any Thoughts about the data?

- What data is repeated over and over again?

CD			
CDNumber	Band	Album	DateReleased
43279	Beatles	White Album	11/22/1968

Track			
TrackNumber	Tune	Composer	DateRecorded
1	Back in the USSR	McCartney	10/7/1968
2	Dear Prudence	Lennon	10/15/1968
3	Glass Onion	Lennon	11/2/1968
4	Ob-La-Di, Ob-La-Da	McCartney	10/9/1968
5	Wild Honey Pie	McCartney	10/25/1968
6	Bungalow Bill	Lennon	10/1/1968
7	While My Guitar Gently Weeps	Harrison	10/20/1968
8	Happiness is a Warm Gun	Lennon	11/4/1968

Musician
John Lennon
Paul McCartney
George Harrison
Ringo Starr

Is this better? Why?

Is it enough?

Issue #1: Where are the relationships?

How does a track know which CD it belongs to?

There's no CDNumber in Track

The relationship exists visually, but not structurally (we need foreign keys to join tables in relational databases)

Album-level data is no longer repeated per track
 (Band, Album, DateReleased live in CD)
Track-level data lives with tracks
 (Tune, Composer, DateRecorded live in Track)
Band members are no longer duplicated across every row

Issue #2: TrackNumber is still context-dependent

TrackNumber is only unique within a CD. If we add more albums it will repeat again.

Issue #3: Composer vs Musician

What if one person composes many tracks?
 What if a track has multiple composers?
 When data represents a real-world entity it usually gets its own table

Issue #4: Musician has no Primary Key

cd			
cdNumber	band	album	dateReleased
43279	1	White Album	11/22/1968

track				
cdNumber	tracknumber	tune	composer	dateRecorded
43279	1	Back in the USSR	2	10/7/1968
43279	2	Dear Prudence	1	10/15/1968
43279	3	Glass Onion	1	11/2/1968
43279	4	Ob-La-Di, Ob-La-Da	2	10/9/1968
43279	5	Wild Honey Pie	2	10/25/1968
43279	6	Bungalow Bill	1	10/1/1968
43279	7	While My Guitar Gently Weeps	3	10/20/1968
43279	8	Happiness is a Warm Gun	1	11/4/1968

band	
bandId	bandName
1	Beatles

band_musician	
bandId	musicianId
1	1
1	2
1	3
1	4

musician	
musicianId	musicianName
1	John Lennon
2	Paul McCartney
3	George Harrison
4	Ringo Starr

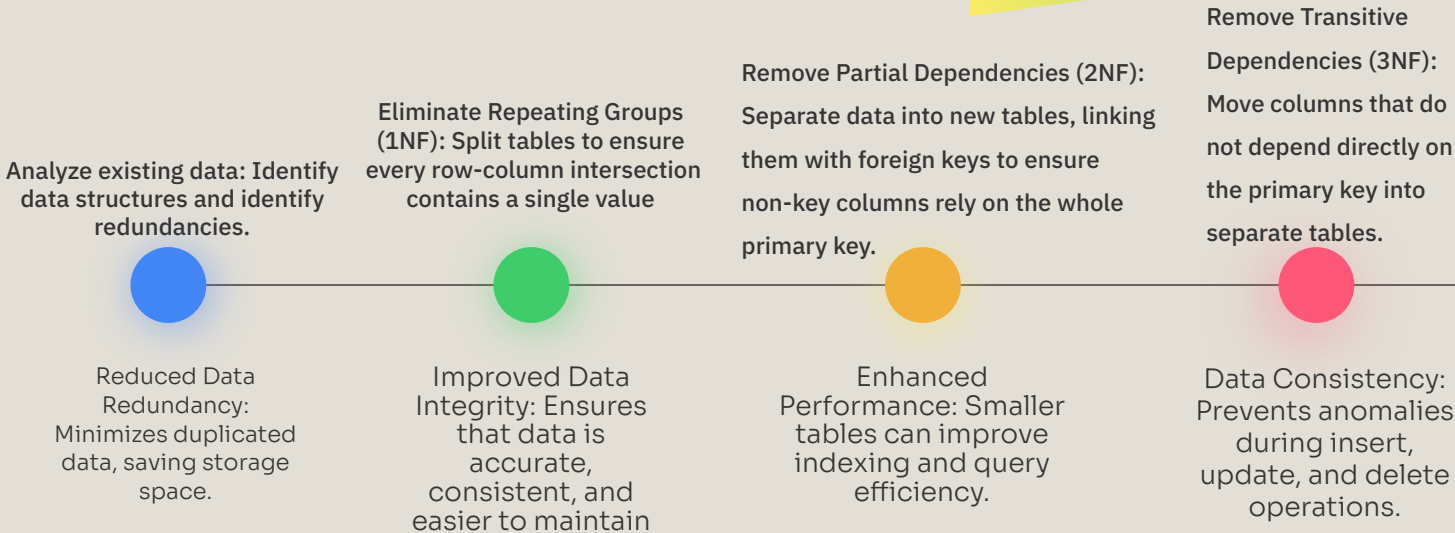
Now if I delete track 8 – “Happiness is a Warm Gun” – what happens?

Suppose The Beatles record a new album, but no tracks are recorded yet?

Suppose we discover the album release date was wrong.

What if Ringo leaves the band?

Normalization is the process of ensuring that database tables:



ANOMALIES

Problem 1

If data is not well structured, anomalies can be introduced:

Problem 2

You may have to perform multiple updates to make one change

Problem 3

You may accidentally delete the wrong data

Problem 4

You may have to add dummy or unnecessary data to insert some specific data

Normalization Process

In order to normalize a database, each table needs to be separately analyzed and moved through a standard process

There are 7 Normal Forms.
(3NF is considered the minimum for a transactional database)

Each Normal Form (NF) is more stringent than the prior one

In order to normalize a table, you must go through a series of steps starting with 1NF and then 2NF etc..

Before normalizing, identify the primary key and determine what depends on it.

Normalization reduces redundancy and anomalies, but may increase the number of joins required in queries. Good design balances integrity and performance

Step 1 - Is the table in 1NF?



A table is in 1NF if it has a primary key and only atomic values:

- It has a Primary Key – Every record has a unique identifier
 - A PK may be constructed using one or more fields
 - There are no repeating groups –
 - Multiple values in one field of a record
 - Multiple fields that are copies of each other
- 

Repeating Groups

customer

customerId	firstName	lastName	email	phone
1	John	Smith	jsmith@gmail.com	2124434434, 2125432222
2	Fred	Jones	fjones@gmail.com	2126563343
3	Tom	Brown	tbrown@gmail.com	3125543644, 7832254462, 3374348838
4	Harry	Baker	hbaker@gmail.com	3125565544

customer

customerId	firstName	lastName	email	phone1	phone2	phone3
1	John	Smith	jsmith@gmail.com	2124434434	2125432222	
2	Fred	Jones	fjones@gmail.com	2126563343		
3	Tom	Brown	tbrown@gmail.com	3125543644	7832254462	3374348838
4	Harry	Baker	hbaker@gmail.com	3125565544		

customer

customerId	firstName	lastName	email
1	John	Smith	jsmith@gmail.com
2	Fred	Jones	fjones@gmail.com
3	Tom	Brown	tbrown@gmail.com
4	Harry	Baker	hbaker@gmail.com

customer_phone

customerId	phone
1	2124434434
1	2125432222
2	2126563343
3	3125543644
3	7832254462
3	3374348838
4	3125565544

course_grade

studentId	courseName	grade	firstName	lastName
1	English	A	John	Smith
2	French	A	Fred	Jones
3	Chinese	B	Tom	Brown
4	English	A	Harry	Baker
1	French	C	John	Smith

Is the table in 1NF?

Yes, primary key and no repeating groups

If so, what is the PK?

studentId + courseName

Is the PK just one field?

Composite

Are all the non-PK fields dependent upon the whole PK?

There is a partial dependency with firstName lastName only dependent on studentId

Normalization: Step 2: Is the table in 2NF?

A table is in 2NF if:

It is in 1NF and no partial dependencies

- non-key attributes are fully functionally dependent on the entire primary key. It eliminates partial dependencies, where columns depend on only part of a composite key, reducing redundancy and update anomalies.
 - **dependent:** A column is dependent on the primary key if the key determines its value.

Move fields that are not dependent on the whole PK to another table

course_grade

studentId (PK)	courseName (PK)	grade	firstName	lastName
1	English	A	John	Smith
2	French	A	Fred	Jones
3	Chinese	B	Tom	Brown
4	English	A	Harry	Baker
1	French	C	John	Smith



course_grade

studentId (PK)	courseName (PK)	grade
1	English	A
2	French	A
3	Chinese	B
4	English	A
1	French	C

Grade depends on both the courseName and the studentId (a student gets a grade in a course)

student

studentId (PK)	firstName	lastName
1	John	Smith
2	Fred	Jones
3	Tom	Brown
4	Harry	Baker

firstName lastName is only dependent on the studentId since many students can be in a course

This table is in 1NF with a composite Primary Key since both studentID and courseName repeat since many students can take many courses

tune

tuneId	tuneTitle	releaseDate	artistId	artistname	publisher
1	Hey Jude	6/1/1967	1	Beatles	Apple
2	Gimme Shelter	7/1/1968	2	Rolling Stones	Atlantic
3	Wish You Were Here	10/1/1978	3	Pink Floyd	United
4	Blackbird	6/1/1968	1	Beatles	RCA
5	Space Oddity	4/1/1971	4	David Bowie	Atlantic

Is the table in 1NF?

Yes, primary key and no repeating groups

If so, what is the PK?

tuneId

Is the table in 2NF?

Yes, there can't be a partial dependency because there is just one PK

Are any of the non PK fields dependent upon other non-PK fields?

Yes, artistName is dependent on artistId so artistName is only dependent on the PK through artistId. That's a transitive dependency.

Normalization: Step 3: Is the table in 3NF?

A table is in 3NF if:

It is in 1NF and 2NF and there are no **transitive dependencies**

- A transitive dependency exists when a non-key attribute depends on another non-key attribute instead of directly on the primary key
- A transitive dependency happens when a column depends on something that depends on the primary key.
- $PK \rightarrow A \rightarrow B$

tune					
tunelid (PK)	tuneTitle	releaseDate	artistId	artistname	publisher
1	Hey Jude	6/1/1967	1	Beatles	Apple
2	Gimme Shelter	7/1/1968	2	Rolling Stones	Atlantic
3	Wish You Were Here	10/1/1978	3	Pink Floyd	United
4	Blackbird	6/1/1968	1	Beatles	RCA
5	Space Oddity	4/1/1971	4	David Bowie	Atlantic

tune					
tunelid (PK)	tuneTitle	releaseDate	artistId	publisher	
1	Hey Jude	6/1/1967	1	Apple	
2	Gimme Shelter	7/1/1968	2	Atlantic	
3	Wish You Were Here	10/1/1978	3	United	
4	Blackbird	6/1/1968	1	RCA	
5	Space Oddity	4/1/1971	4	Atlantic	

artist	
artistId (PK)	artistname
1	Beatles
2	Rolling Stones
3	Pink Floyd
4	David Bowie

A transitive dependency exists here because artistName depends on artistId, not directly on tunelid.

$PK \rightarrow A \rightarrow B$. If an artist changed their name, we'd have to update multiple rows, that's redundancy and an update anomaly

Last step:

Create lookup tables for repeated categorical values

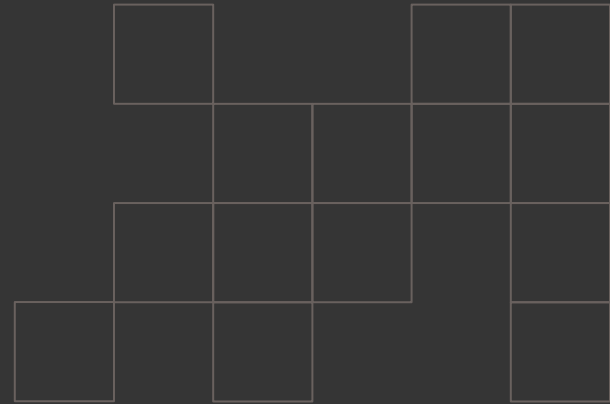
student			
studentId	firstName	lastName	status
1	Tom	Brown	Active
2	John	Smith	Pending
3	Fred	Jones	Graduated
4	Dick	Tracy	Active
5	Sally	Smith	Active
6	Mary	Baker	Pending
7	Harry	Taylor	Active
8	Jerry	Jones	Active
9	Susan	Chambers	Graduated
10	Julie	Brown	Active



student			
studentId	firstName	lastName	statusId
1	Tom	Brown	1
2	John	Smith	2
3	Fred	Jones	3
4	Dick	Tracy	1
5	Sally	Smith	1
6	Mary	Baker	2
7	Harry	Taylor	1
8	Jerry	Jones	1
9	Susan	Chambers	3
10	Julie	Brown	1

status	
statusId	StatusName
1	Active
2	Pending
3	Graduated

Book Normalization Example



book

bookTitle	authorNumber	lastName	publisher	royalty	edition
Introduction to programming	453	Williams	Collins	25	4
Introduction to programming	453	Williams	Collins	29	5
SQL Explained	235	Smith	Collins	34	4
Normalization	255	Brown	Pearson	23	2
Database Theory	634	Turner	Wilkie	22	3
Object Oriented Programming	453	Williams	Pearson	14	5
Oracle for Beginners	223	Baker	Collins	25	3
Primary keys	235	Smith	Cengage	32	1
What is a table	523	Jones	Pearson	23	5
Relations Explained	244	Robinson	Persea	42	3
Python Programming	523	Jones	Hyperion	14	2

Does this table have a primary key? If so, what is it?

No, we could make a composite bookTitle, edition, authorNumber but we should probably just add a bookId

Is this table in a Normal Form? If so, which?

This table as is not in any normal form because there is no primary key and it has repeating values so it fails even 1NF

book_edition

<u>bookTitle</u>	<u>edition</u>	royalty
Introduction to programming	4	25
Introduction to programming	5	29
SQL Explained	4	34
Normalization	2	23
Database Theory	3	22
Object Oriented Programming	5	14
Oracle for Beginners	3	25
Primary keys	1	32
What is a table	5	23
Relations Explained	3	42
Python Programming	2	14

Does this table have a primary key? If so, what is it?

Yes, bookTitle + edition is a composite PK

Is this table in a Normal Form? If so, which?

Yes, this table is in 3NF. No partial or transitive dependencies

book

bookTitle	authorNumber	lastName	publisher
Introduction to programming	453	Williams	Collins
SQL Explained	235	Smith	Collins
Normalization	255	Brown	Pearson
Database Theory	634	Turner	Wilkie
Object Oriented Programming	453	Williams	Pearson
Oracle for Beginners	223	Baker	Collins
Primary keys	235	Smith	Cengage
What is a table	523	Jones	Pearson
Relations Explained	244	Robinson	Persea
Python Programming	523	Jones	Hyperion

Does this table have a primary key? If so, what is it?

We could say bookTitle is

Is this table in a Normal Form? If so, which?

This table is in 2NF but not 3NF because there is a transitive dependency : bookTitle → authorNumber → lastName

book_edition

<u>bookTitle</u>	<u>edition</u>	royalty
Introduction to programming	4	25
Introduction to programming	5	29
SQL Explained	4	34
Normalization	2	23
Database Theory	3	22
Object Oriented Programming	5	14
Oracle for Beginners	3	25
Primary keys	1	32
What is a table	5	23
Relations Explained	3	42
Python Programming	2	14

book

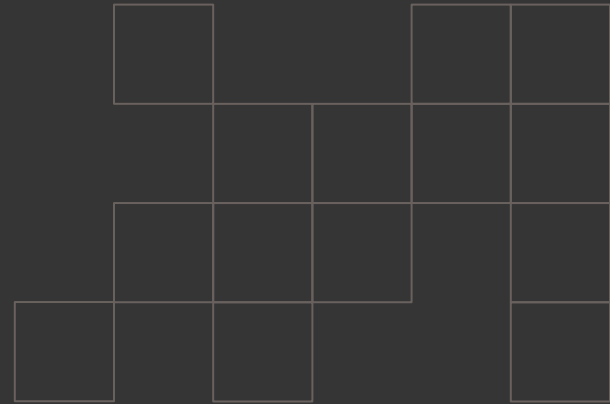
<u>bookTitle</u>	authorNumber	publisher
Introduction to programming	453	Collins
SQL Explained	235	Collins
Normalization	255	Pearson
Database Theory	634	Wilkie
Object Oriented Programming	453	Pearson
Oracle for Beginners	223	Collins
Primary keys	235	Cengage
What is a table	523	Pearson
Relations Explained	244	Persea
Python Programming	523	Hyperion

Is this table in a Normal Form? If so, which?
Yes, these tables are in 3NF.

author

<u>authorNumber</u>	lastName
223	Baker
235	Smith
244	Robinson
255	Brown
453	Williams
523	Jones
634	Turner

Project Normalization Example



project

projectNumber	projectName	employeeNumber	employeeName	jobType	hourlyRate	hoursWorked
15	Evergreen	103	June	Engineer	84	32
15	Evergreen	101	John	Database Designer	105	23
15	Evergreen	105	Alice	Application Designer	100	43
15	Evergreen	106	William	Programmer	56	23
15	Evergreen	102	David	Systems Analyst	95	12
22	Amber	105	Alice	Application Designer	100	34
22	Amber	104	Anne	Programmer	56	12
22	Amber	113	Edward	Clerical Support	33	32
22	Amber	111	Sarah	Programmer	56	21
22	Amber	106	William	Programmer	56	31
25	Starlight	107	Fred	Systems Analyst	95	14
25	Starlight	115	Maria	Programmer	56	15
25	Starlight	101	John	Database Designer	105	13
25	Starlight	114	Anne	Quality Assurance	63	23
25	Starlight	113	Edward	Clerical Support	33	21

Does this table have a primary key? If so, what is it?

Is this table in a normal form? If so, what is it?

Composite PK = (projectNumber, employeeNumber)

1NF?

Yes — all fields are atomic.

So it passes 1NF

assignment

<u>projectNumber</u>	<u>employeeNumber</u>	hoursWorked
15	103	32
15	101	23
15	105	43
15	106	23
15	102	12
22	105	34
22	104	12
22	113	32
22	111	21
22	106	31
25	107	14
25	115	15
25	101	13
25	114	23
25	113	21

project

<u>projectNumber</u>	projectName
15	Evergreen
22	Amber
25	Starlight

employee

employeeNumber	employeeName	jobType	hourlyRate
101	John	Database Designer	105
102	David	Systems Analyst	95
103	June	Engineer	84
104	Anne	Programmer	56
105	Alice	Application Designer	100
106	William	Programmer	56
107	Fred	Systems Analyst	95
111	Sarah	Programmer	56
113	Edward	Clerical Support	33
114	Anne	Quality Assurance	63
115	Maria	Programmer	56

Does this table have a primary key? If so, what is it?

Yes, employeeNumber

Is this table in a normal form? If so, what is it?

It is 2NF but not 3NF because hourlyRate is transitively dependet on the PK through jobType. employeeNumber determines jobType, and jobType determines hourlyRate — so hourlyRate depends on the primary key indirectly (a transitive dependency).

assignment

<u>projectNumber</u>	<u>employeeNumber</u>	hoursWorked
15	103	32
15	101	23
15	105	43
15	106	23
15	102	12
22	105	34
22	104	12
22	113	32
22	111	21
22	106	31
25	107	14
25	115	15
25	101	13
25	114	23
25	113	21

project

<u>projectNumber</u>	projectName
15	Evergreen
22	Amber
25	Starlight

job_type

<u>jobType</u>	hourlyRate
Application Designer	100
Clerical Support	33
Database Designer	105
Engineer	84
Programmer	56
Quality Assurance	63
Systems Analyst	95

employee

<u>employeeNumber</u>	employeeName	jobType
101	John	Database Designer
102	David	Systems Analyst
103	June	Engineer
104	Anne	Programmer
105	Alice	Application Designer
106	William	Programmer
107	Fred	Systems Analyst
111	Sarah	Programmer
113	Edward	Clerical Support
114	Anne	Quality Assurance
115	Maria	Programmer

These tables are all
normalized to 3NF